MSG No. 138.521 $P4_2/n'cm$ [Type III, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,4,11,12]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,3,9,10]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[5,6,14,15]`
4	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[7,8,13,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,4,14,15]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,13,16]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[5,6,11,12]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[7,8,9,10]`

Table 3: Wyckoff site: `4c`, site symmetry: `..2'/m`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,8,12,13]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,5,9,14]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[3,6,10,15]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,7,11,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `..2'/m`

No.	position	mapping
1	$[0, 0, 0]$	`[1,8,12,13]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,5,9,14]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,6,10,15]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,7,11,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,4,7,8]`
2	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[2,3,5,6]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[9,10,14,15]`
4	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[11,12,13,16]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,4]`
2	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[2,3]`
3	$\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[5,6]`
4	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	`[7,8]`
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[9,10]`
6	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	`[11,12]`
7	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[13,16]`
8	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[14,15]`

Table 7: Wyckoff site: `8g`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x, -x, \frac{1}{2}\right]$	`[1,12]`
2	$\left[x+\frac{1}{2}, x, 0\right]$	`[2,9]`
3	$\left[-x, \frac{1}{2}-x, 0\right]$	`[3,10]`
4	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}\right]$	`[4,11]`
5	$\left[\frac{1}{2}-x, -x, 0\right]$	`[5,14]`
6	$\left[x, x+\frac{1}{2}, 0\right]$	`[6,15]`
7	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	`[7,16]`
8	$\left[-x, x, \frac{1}{2}\right]$	`[8,13]`

Table 8: Wyckoff site: `8h`, site symmetry: `..2'`

No.	position	mapping
1	$[x, -x, 0]$	`[1,12]`
2	$\left[x+\frac{1}{2}, x, \frac{1}{2}\right]$	`[2,9]`
3	$\left[-x, \frac{1}{2}-x, \frac{1}{2}\right]$	`[3,10]`
4	$\left[\frac{1}{2}-x, x+\frac{1}{2}, 0\right]$	`[4,11]`
5	$\left[\frac{1}{2}-x, -x, \frac{1}{2}\right]$	`[5,14]`
6	$\left[x, x+\frac{1}{2}, \frac{1}{2}\right]$	`[6,15]`
7	$\left[x+\frac{1}{2}, \frac{1}{2}-x, 0\right]$	`[7,16]`
8	$[-x, x, 0]$	`[8,13]`

Table 9: Wyckoff site: `8i`, site symmetry: `..m`

No.	position	mapping
1	$[x, x, z]$	`[1,8]`
2	$\left[\frac{1}{2}-x, x, z+\frac{1}{2}\right]$	`[2,5]`
3	$\left[x, \frac{1}{2}-x, z+\frac{1}{2}\right]$	`[3,6]`
4	$\left[\frac{1}{2}-x, \frac{1}{2}-x, z\right]$	`[4,7]`
5	$\left[x+\frac{1}{2}, -x, \frac{1}{2}-z\right]$	`[9,14]`
6	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-z\right]$	`[10,15]`
7	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -z\right]$	`[11,16]`

continued ...

Table 9

No.	position	mapping
8	$[-x,\, -x,\, -z]$	`[12,13]`

Table 10: Wyckoff site: `16j`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[\frac{1}{2}-y,\, x,\, z+\frac{1}{2}\right]$	`[2]`
3	$\left[y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[3]`
4	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	`[4]`
5	$\left[\frac{1}{2}-x,\, y,\, z+\frac{1}{2}\right]$	`[5]`
6	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	`[6]`
7	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	`[7]`
8	$[y,\, x,\, z]$	`[8]`
9	$\left[x+\frac{1}{2},\, -y,\, \frac{1}{2}-z\right]$	`[9]`
10	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[10]`
11	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	`[11]`
12	$[-y,\, -x,\, -z]$	`[12]`
13	$[-x,\, -y,\, -z]$	`[13]`
14	$\left[y+\frac{1}{2},\, -x,\, \frac{1}{2}-z\right]$	`[14]`
15	$\left[-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[15]`
16	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	`[16]`